1. An ammeter is connected in **series** with the load.

2. A voltmeter is connected in **parallel** with the load.

3. Name two types of PMMC galvanometer meter movements:**D'Arsonval Movement& Taut Band Movement**

4. The same basic electrical principles govern the behavior of the PMMC meters and **electrodynamometer meters.**

5. The range of an ammeter, milliammeter, or microammeter can be increased if a **shunt resistor** is used in parallel with the meter movement

6. A voltmeter can be made from a DC current meter movement if a **multiplier resistor** is connected in series with the meter movement.

7. Draw the circuit diagram for a simple DC voltmeter:

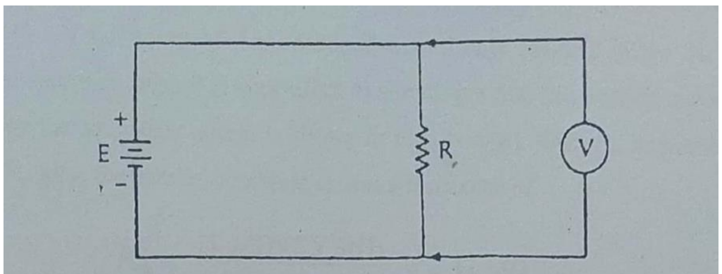


Fig. (1-9) A voltmeter is always connected in parallel with the load.

8. A current meter should have an internal resistance that is **low** compared with external circuit resistances.

9. A voltmeter should have an internal resistance that is **high** compared with external circuit resistances.

10. An ohmmeter may be used in circuits with the power turned off. **(True)**

11. A thermocouple AC ammeter reads the **RMS** value of the AC waveform.

12. Thermocouple ammeters are used at frequencies up to **100 MHz.**

13. A hot-wire ammeter reads the **RMS** value of the AC waveform.

14. A dynamometer is an AC ammeter constructed from a movable coil in the magnetic fields of two stationary coils. All three coils are connected in **series** with each other.

15. The ammeter in Question 14 is calibrated in RMS values but is actually reading **average** values.

16. A **current transformer** is used to extend the current range of the electrodynamometer type of AC current meter.

17. Iron-vane meters read the **RMS** value of the AC waveform.

18. What happens to the iron-vane meter when DC is measured?

The meter suffers from residual magnetism due to the unidirectional current.

Method to overcome this problem: Use a demagnetizing circuit or alternating current before measuring DC.

19. A rectifier meter reads the **average** value of the alternating current or AC voltage waveform.

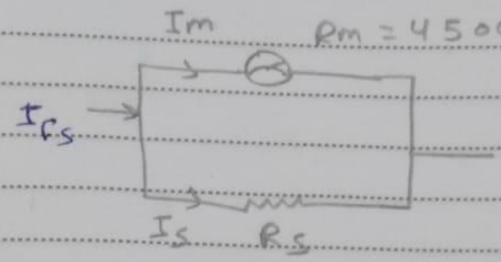
20. A diode is often used as the instrumentation rectifier in audio frequency AC meters.**(True)**

21. An AC meter reads different values depending on which way the meter probes are connected. This is called **turnover effect.**

22. A rectifier-type AC voltmeter is operated such that the voltage applied to the rectifier diode is less than 0.7 V (for silicon diodes). This is an example of **square-law** device.

23. Which type of AC meter rectifier is immune to both turnover effect and the effects of the phase of harmonics on the reading? **Full-wave square-law rectifier.**

1-



$I_m = 50 \mu A$, $R_m = 4500$, $I_{FS} = 500 \mu A$

∴ $I_s = I_{fs} - I_m = 450 \mu A$

$$I_S = I_{fs} \frac{R_m}{R_m + R_S}$$

$$450 = 500 \times \frac{4500}{4500 + R_S}$$

∴ $R_S + 4500 = 5000$

$R_S = 500 \Omega$

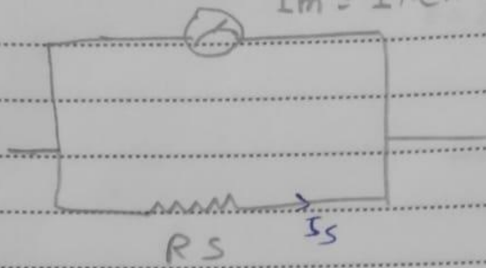
2- $I_m = \frac{E}{R_1} = \frac{9}{12} = .75$ A

3- $I_m = \frac{E}{R_S} = \frac{12}{22} = .54$ A

Source

$I_m = 1\mu A \quad R_m = 1100$

4 -



$I_{FS} = 100\mu$

a) ohm's law :-

$I_S = I_{FS} - I_M$
$= 100 - 1 = 99 \mu A$

$E_M = I_m R_m = 10^{-6} \times 1100 = 1{,}1 \, mV$

$\therefore R_S = \frac{E_M}{I_S} = \frac{1{,}1 \times 10^{-3}}{99 \times 10^{-6}} = 11{,}1$

b) current divider :-

$I_S = I_{FS} \frac{R_m}{R_S + R_m}$

$99 = 100 \times \frac{1100}{1100 + R_S}$

$1100 + R_S = 1111{,}11 \qquad \therefore R_S = 11{,}1 \, \Omega$

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6- $I_m = \dfrac{E}{R_m + R_{mx}}$

$100 \times 10^{-6} = \dfrac{100}{1300 + R_{mx}}$

$\therefore R_{mx} = 998700\ \Omega$

7- $100 \times 10^{-6} = \dfrac{2}{1300 + R_{mx}}$

$R_{mx} = 18700\ \Omega$

8- $\phi = \dfrac{1}{I_{fs}} = \dfrac{1}{30 \times 10^{-6}} = \dfrac{1}{30} M\ \Omega / V$

9- $R_M = E_{Fs}\, \phi$

$= 50 \times 20{,}000 = 1 M\Omega$